

1	2	3	4
+	+	+	+

Теория вероятностей
Индивидуальное домашнее задание
Вариант 11

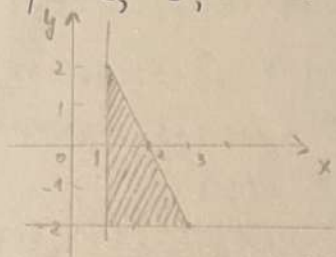
Группа: 5130201/30101

Студент: Мартыненко Анна Павловна

Проверил: Малов Сергей Васильевич

$$\begin{cases} 4x+2y \leq 8 \\ x \geq 1, y \geq -2 \end{cases}$$

$$\xi = -2\eta^2 - 3, \eta = [-3; 7], \mu = 12\xi + 6\eta$$



$$1) p_{\xi, \eta}(x, y) = \begin{cases} c, & x \in S \\ 0, & \text{otherwise} \end{cases}, \text{ где } S = \begin{cases} 4x+2y \leq 8 \\ x \geq 1 \\ y \geq -2 \end{cases}$$

$$\iint_{\mathbb{R}^2} p_{\xi, \eta}(x, y) dx dy = 1$$

$$c \iint_S dx dy = c \int_1^3 dx \int_{-2}^{4-2x} dy = c \int_1^3 dx y \Big|_{-2}^{4-2x} = c \int_1^3 (4-2x+2) dx = c \int_1^3 (-2x+6) dx = c \left(-\frac{2x^2}{2} + 6x \right) \Big|_1^3 =$$

$$= c(-9+18+1-6) = 4c = 1 \Rightarrow c = \frac{1}{4}$$

$$p_{\xi, \eta} = \begin{cases} \frac{1}{4}, & x \in S \\ 0, & \text{otherwise} \end{cases}$$

$$p_{\xi}(x) = \iint_{\mathbb{R}^2} p_{\xi, \eta}(x, y) dy = \int_{-2}^{4-2x} \frac{1}{4} dy = \frac{1}{4} y \Big|_{-2}^{4-2x} = \frac{1}{4} (-2x+4+2) = \frac{1}{4} (-2x+6) = \frac{3-x}{2}, \quad x \in [1; 3]$$

$$p_{\eta}(y) = \iint_{\mathbb{R}^2} p_{\xi, \eta}(x, y) dx = \int_1^{2-\frac{1}{2}y} \frac{1}{4} dx = \frac{1}{4} x \Big|_1^{2-\frac{1}{2}y} = \frac{1}{4} \left(-\frac{1}{2}y + 2 - 1 \right) = \frac{1}{4} \left(-\frac{1}{2}y + 1 \right) = \frac{2-y}{8}, \quad y \in [-2; 2]$$

$$F_{\xi}(x) = \int_1^x \frac{3-t}{2} dt = \left(\frac{3}{2}t - \frac{t^2}{4} \right) \Big|_1^x = \left(\frac{3}{2}x - \frac{3}{2} - \frac{x^2}{4} + \frac{1}{4} \right) = -\frac{x^2}{4} + \frac{3}{2}x - \frac{5}{4}, \quad x \in [1; 3]$$

$$F_{\eta}(y) = \int_{-2}^y \frac{2-t}{8} dt = \frac{1}{8} \left(2t - \frac{t^2}{2} \right) \Big|_{-2}^y = \frac{1}{8} \left(2y - \frac{y^2}{2} - (-4 - 2) \right) = -\frac{y^2}{16} + \frac{1}{4}y + \frac{3}{4}, \quad y \in [-2; 2]$$

$$p_{\xi}(x) = \begin{cases} \frac{3-x}{2}, & x \in [1; 3] \\ 0, & \text{otherwise} \end{cases}$$

$$F_{\xi}(x) = \begin{cases} 0, & x \leq 1 \\ -\frac{x^2}{4} + \frac{3}{2}x - \frac{5}{4}, & x \in [1; 3] \\ 1, & x > 3 \end{cases}$$

$$p_{\eta}(y) = \begin{cases} \frac{2-y}{8}, & y \in [-2; 2] \\ 0, & \text{otherwise} \end{cases}$$

$$F_{\eta}(y) = \begin{cases} 0, & y \leq -2 \\ -\frac{y^2}{16} + \frac{1}{4}y + \frac{3}{4}, & y \in [-2; 2] \\ 1, & y > 2 \end{cases}$$

проверим независимость компонент:

$$\exists x=1, y=1$$

$$p_{\xi}(x=1) = \frac{3-1}{2} = 1$$

$$p_{\eta}(y=1) = \frac{2-1}{8} = \frac{1}{8}$$

$$\Rightarrow p_{\xi} \cdot p_{\eta} = 1 \cdot \frac{1}{8} = \frac{1}{8} \neq p_{\xi, \eta}(1, 1) = \frac{1}{4}$$

$p_{\xi, \eta} \neq p_{\xi} \cdot p_{\eta} \Rightarrow$ величины зависимы

проверка корректности функций и плотностей распределения компонент:

$$\int_1^3 \frac{3-x}{2} dx = \frac{1}{2} \left(3x - \frac{x^2}{2} \right) \Big|_1^3 = \frac{1}{2} \left(9 - \frac{9}{2} - \left(3 - \frac{1}{2} \right) \right) = \frac{9}{2} - \frac{3}{4} - \left(\frac{3}{2} - \frac{1}{4} \right) = 1 \Rightarrow \text{верно}$$

$$F_{\xi}(1) = -\frac{1}{4} + \frac{3}{2} - \frac{5}{4} = 0$$

$\Rightarrow$ верно

$$F_{\xi}(3) = -\frac{9}{4} + \frac{9}{2} - \frac{5}{4} = \frac{4}{4} = 1$$

$$\int_{-2}^2 \frac{2-y}{8} dy = \frac{1}{8} \left(2y - \frac{y^2}{2} \right) \Big|_{-2}^2 = \left(\frac{1}{4}y - \frac{y^2}{16} \right) \Big|_{-2}^2 = \frac{2}{4} - \frac{4}{16} - \left(-\frac{2}{4} - \frac{4}{16} \right) = 1 \Rightarrow \text{верно}$$

$$F_{\eta}(-2) = -\frac{4}{16} + \frac{-2}{4} + \frac{3}{4} = 0$$

$\Rightarrow$ верно

$$F_{\eta}(2) = -\frac{4}{16} + \frac{2}{4} + \frac{3}{4} = \frac{4}{4} = 1$$

$$a) G = -2\xi^2 - 3$$

$$\text{supp } \xi = [1, 3]$$

$$\text{supp } G = [-21, -5]$$

$$F_G(x) = P(G < x) = P(-2\xi^2 - 3 < x) = P(-2\xi^2 < x + 3) = P(\xi^2 > \frac{-x-3}{2}) = P(\xi > \sqrt{\frac{-x-3}{2}}) = 1 - F_\xi \sqrt{\frac{-x-3}{2}}$$

$$x \in [-21, -5]: F_G = 1 - \left(-\frac{1}{4} \left(\frac{-x-3}{2} \right) + \frac{3}{2} \sqrt{\frac{-x-3}{2}} - \frac{5}{4} \right) = \frac{15-x}{8} - \frac{3\sqrt{-2x-6}}{4}$$

$$\text{проверка: } F_G(-21) = \frac{15+21}{8} - \frac{3\sqrt{42-6}}{4} = \frac{36}{8} - \frac{18}{4} = 0$$

$$F_G(-5) = \frac{15+5}{8} - \frac{3\sqrt{10-6}}{4} = \frac{20}{8} - \frac{12}{8} = \frac{8}{8} = 1 \Rightarrow \text{верно}$$

$$\left(\frac{15-x}{8} - \frac{3\sqrt{-2x-6}}{4} \right)' = -\frac{1}{8} + \frac{3}{4\sqrt{2}\sqrt{-x-3}}$$

$$\text{проверка: } \int_{-21}^{-5} -\frac{1}{8} + \frac{3}{4\sqrt{2}\sqrt{-x-3}} dx = -\frac{1}{8}x \Big|_{-21}^{-5} + \frac{3}{4\sqrt{2}} \sqrt{-2x-6} \Big|_{-21}^{-5} = -\frac{1}{8}(-5) - \frac{3}{4}\sqrt{-2(-5)-6} - \left(-\frac{1}{8}(-21) - \frac{3}{4}\sqrt{-2(-21)-6} \right) = 1 \Rightarrow \text{верно}$$

$$F_G(x) = \begin{cases} 0, & x \leq -21 \\ \frac{15-x}{8} - \frac{3\sqrt{-2x-6}}{4}, & x \in [-21, -5] \\ 1, & x > -5 \end{cases}$$

$$p_G(x) = \begin{cases} -\frac{1}{8} + \frac{3}{4\sqrt{2}\sqrt{-x-3}}, & x \in [-21, -5] \\ 0, & \text{otherwise} \end{cases}$$

$$EG = \int_{-\infty}^{\infty} x p_G(x) dx = \int_{-21}^{-5} x \left(-\frac{1}{8} + \frac{3}{4\sqrt{2}\sqrt{-x-3}} \right) dx = \int_{-21}^{-5} -\frac{x}{8} + \frac{3x}{4\sqrt{2}\sqrt{-x-3}} dx = -\frac{x^2}{16} \Big|_{-21}^{-5} - \left(\sqrt{-x-3} \left(\frac{x}{2\sqrt{2}} - \frac{3}{\sqrt{2}} \right) \right) \Big|_{-21}^{-5} = \frac{63}{16} - \frac{207}{16} = -\frac{144}{16} = -9$$

$$EG^2 = \int_{-\infty}^{\infty} x^2 p_G(x) dx = \int_{-21}^{-5} -\frac{x^2}{8} + \frac{3x^2}{4\sqrt{2}\sqrt{-x-3}} dx = \left(-\frac{x^3}{24} - 3\sqrt{-x-3} \left(\frac{x^2}{5\sqrt{2}} - \frac{\sqrt{2}x}{5} + \frac{3\sqrt{2}}{5} \right) \right) \Big|_{-21}^{-5} = \frac{1391}{15} \approx 92.73$$

$$D_G = EG^2 - (EG)^2 = 92.73 - 81 \approx 11.73$$

$$b) D = [3\eta]$$

$$\text{supp } \eta = [-2, 2] \quad \text{supp } D = \{-6, -5, \dots, 5, 6\}$$

$$P(D=k) = P(\eta \in \left(\frac{k}{3}, \frac{k+1}{3} \right)) = \int_{k/3}^{(k+1)/3} \frac{2-y}{8} dy = \left(\frac{y}{4} - \frac{y^2}{16} \right) \Big|_{\frac{k}{3}}^{\frac{k+1}{3}}$$

D	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	Σ
P	$\frac{23}{144}$	$\frac{7}{48}$	$\frac{19}{144}$	$\frac{17}{144}$	$\frac{5}{48}$	$\frac{13}{144}$	$\frac{11}{144}$	$\frac{1}{16}$	$\frac{7}{144}$	$\frac{5}{144}$	$\frac{1}{48}$	$\frac{1}{144}$	0	1

$$\text{пример вычисления для } D = -6: P(D = -6) = \left(\frac{y}{4} - \frac{y^2}{16} \right) \Big|_{-2}^{-5/3} = \frac{-5/3+2}{4} - \frac{(-5/3)^2-4}{16} = \frac{23}{144}$$

$$ED = (-6) \cdot \frac{23}{144} + (-5) \cdot \frac{7}{48} + \dots + 5 \cdot \frac{1}{144} \approx -\frac{179}{42} \approx -2.48$$

$$ED^2 = (-6)^2 \cdot \frac{23}{144} + (-5)^2 \cdot \frac{7}{48} + \dots + 5^2 \cdot \frac{1}{144} = \frac{1019}{72}$$

$$D_D = ED^2 - (ED)^2 = \frac{1019}{72} - \left(-\frac{179}{42} \right)^2 \approx 7.97$$

$$3) E\xi = \int_{-\infty}^{\infty} x p_\xi(x) dx = \int_1^3 x \frac{3-x}{2} dx = \left(\frac{3x^2}{4} - \frac{x^3}{6} \right) \Big|_1^3 = \frac{27}{4} - \frac{27}{6} - \left(\frac{3}{4} - \frac{1}{6} \right) = \frac{5}{3}$$

$$E\xi^2 = \int_{-\infty}^{\infty} x^2 p_\xi(x) dx = \int_1^3 x^2 \frac{3-x}{2} dx = \left(\frac{x^3}{2} - \frac{x^4}{8} \right) \Big|_1^3 = \frac{27}{2} - \frac{81}{8} - \left(\frac{1}{2} - \frac{1}{8} \right) = 3$$

$$D_\xi = E\xi^2 - (E\xi)^2 = 3 - \left(\frac{5}{3} \right)^2 = \frac{2}{9}$$

$$E\eta = \int_{-\infty}^{\infty} y p_\eta(y) dy = \int_{-2}^2 y \frac{2-y}{8} dy = \left(\frac{y^2}{8} - \frac{y^3}{24} \right) \Big|_{-2}^2 = \frac{4}{8} - \frac{8}{24} - \frac{4}{8} + \frac{8}{24} = -\frac{2}{3}$$

$$E\eta^2 = \int_{-\infty}^{\infty} y^2 p_\eta(y) dy = \int_{-2}^2 y^2 \frac{2-y}{8} dy = \left(\frac{y^3}{12} - \frac{y^4}{32} \right) \Big|_{-2}^2 = \frac{8}{12} - \frac{16}{32} + \frac{8}{12} - \frac{16}{32} = \frac{4}{3}$$

$$D_\eta = E\eta^2 - (E\eta)^2 = \frac{4}{3} - \left(-\frac{2}{3} \right)^2 = \frac{8}{9}$$

$$E\left(\frac{\xi}{\eta} \right) = \begin{pmatrix} 5/3 \\ -2/3 \end{pmatrix}$$

$$\begin{aligned} E_{\xi\eta} &= \iint_{R^2} xy p_{\xi\eta}(x,y) dx dy = \int_1^3 dx \int_{-2}^{-2x+4} xy \frac{1}{4} dy = \frac{1}{4} \int_1^3 x dx \left[\frac{y^2}{2} \right]_{-2}^{-2x+4} = \frac{1}{4} \int_1^3 x \left(\frac{(-2x+4)^2}{2} - 4 \right) dx = \\ &= \frac{1}{8} \int_1^3 x (4x^2 - 16x + 12) dx = \frac{1}{8} \int_1^3 (4x^3 - 16x^2 + 12x) dx = \frac{1}{8} \left(x^4 - \frac{16x^3}{3} + \frac{12x^2}{2} \right) \Big|_1^3 = \\ &= \frac{1}{8} \left(81 - \frac{16 \cdot 27}{3} + \frac{12 \cdot 9}{2} - \left(1 - \frac{16}{3} + \frac{12}{2} \right) \right) = -\frac{4}{3} \end{aligned}$$

$$\text{cov}(\xi, \eta) = E_{\xi\eta} - E_{\xi} \cdot E_{\eta} = -\frac{4}{3} + \frac{5}{3} \cdot \frac{2}{3} = -\frac{2}{9}$$

$$\text{var} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} D_{\xi} & \text{cov}(\xi, \eta) \\ \text{cov}(\xi, \eta) & D_{\eta} \end{pmatrix} = \begin{pmatrix} \frac{2}{9} & -\frac{2}{9} \\ -\frac{2}{9} & \frac{8}{9} \end{pmatrix}$$

$$\rho(\xi, \eta) = \frac{\text{cov}(\xi, \eta)}{\sqrt{D_{\xi} \cdot D_{\eta}}} = \frac{-\frac{2}{9}}{\sqrt{\frac{2}{9} \cdot \frac{8}{9}}} = \frac{-\frac{2}{9}}{\frac{2\sqrt{2}}{9}} = \frac{-2}{2\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

$$p_{\xi|\eta}(x) = \frac{p_{\xi\eta}(x,y)}{p_{\eta}(y)} = \frac{1 \cdot 8}{4 \cdot (2-y)} = \frac{2}{2-y} \text{ при } x \in [1, \frac{4-y}{2}]$$

$$p_{\xi|\eta}(x) = \begin{cases} \frac{2}{2-y}, & x \in [1, \frac{4-y}{2}] \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} E(\xi|\eta=y) &= \int_1^{\frac{4-y}{2}} \frac{2}{2-y} \cdot x dx = \frac{2}{2-y} \left[\frac{x^2}{2} \right]_1^{\frac{4-y}{2}} = \frac{2}{2-y} \left(\frac{(4-y)^2}{4} - \frac{1}{2} \right) = \frac{2}{2-y} \frac{(4-y)^2 - 2}{4} = \frac{16 - 8y + y^2 - 2}{4(2-y)} = \frac{(y-2)(y-6)}{4(2-y)} = \\ &= \frac{6-y}{4} \Rightarrow E(\xi|\eta) = \frac{6-\eta}{4} \end{aligned}$$

$$\begin{aligned} E(\xi^2|\eta=y) &= \int_1^{\frac{4-y}{2}} \frac{2}{2-y} x^2 dx = \frac{2}{2-y} \left[\frac{x^3}{3} \right]_1^{\frac{4-y}{2}} = \frac{2}{2-y} \frac{1}{3} \left(\frac{(4-y)^3}{8} - \frac{1}{3} \right) = \frac{2}{2-y} \frac{(4-y)^3 - 8}{24} = \frac{56 - 48y + 12y^2 - y^3}{12(2-y)} = \\ &= \frac{-(y-2)(y^2 - 10y + 28)}{12(2-y)} = \frac{y^2 - 10y + 28}{12} \end{aligned}$$

$$\begin{aligned} D(\xi|\eta=y) &= E(\xi^2|\eta=y) - (E(\xi|\eta=y))^2 = \frac{y^2 - 10y + 28}{12} - \left(\frac{6-y}{4} \right)^2 = \frac{y^2 - 10y + 28}{12} - \frac{y^2 - 12y + 36}{16} = \\ &= \frac{4y^2 - 40y + 112 - 3y^2 + 36y - 108}{48} = \frac{y^2 - 4y + 4}{48} = \frac{(y-2)^2}{48} \Rightarrow D(\xi|\eta) = \frac{(\eta-2)^2}{48} \end{aligned}$$

4) $\mu = 12\xi + 6\eta$

$\text{supp } \mu = [0, 24]$

$12x + 6y = t$

$12 \cdot 1 + 6 \cdot (-2) = 0$

$12 \cdot 3 + 6 \cdot (-2) = 24$

$t \in [0, 24] : 12x + 6y = t \Rightarrow y = \frac{t - 12x}{6}$

$$\begin{aligned} F_{\mu}(t) &= \int_1^{\frac{12+t}{12}} dx \int_{-\frac{t-12x}{6}}^{\frac{t-12x}{6}} \frac{1}{4} dy = \frac{1}{4} \int_1^{\frac{12+t}{12}} \frac{t - 12x + 12}{6} dx = \frac{1}{24} (tx - 6x^2 + 12x) \Big|_1^{\frac{12+t}{12}} = \\ &= \frac{1}{24} \left(\left(\frac{(12+t)t}{12} - \frac{6(12+t)^2}{144} + \frac{12(12+t)}{12} \right) - (t - 6 + 12) \right) = \frac{1}{24} \left(\frac{12t + t^2}{12} - \frac{144 - 24t - t^2}{24} + 6 \right) = \\ &= \frac{1}{24} \left(\frac{24t + 2t^2 - 144 - 24t - t^2 + 144}{24} \right) = \frac{t^2}{576} \end{aligned}$$

$\left(\frac{t^2}{576} \right)' = \frac{t}{288}$

проверка: $\int_0^{24} \frac{t}{288} dt = \frac{1}{288} \frac{t^2}{2} \Big|_0^{24} = \frac{576}{288 \cdot 2} = 1 \Rightarrow \text{верно}$

$F_{\mu}(0) = 0$

$F_{\mu}(24) = \frac{576}{576} \Rightarrow \text{верно}$

$F_{\mu}(t) = \begin{cases} 0, & t \leq 0 \\ \frac{t^2}{576}, & t \in [0, 24] \\ 1, & t > 24 \end{cases}$

$p_{\mu} = \begin{cases} \frac{t}{288}, & t \in [0, 24] \\ 0, & \text{otherwise} \end{cases}$

$E_{\mu} = \int_{-\infty}^{\infty} t p_{\mu}(t) dt = \int_0^{24} \frac{t^2}{288} dt = \frac{1}{288} \frac{t^3}{3} \Big|_0^{24} = \frac{4608}{288} = 16$

$E_{\mu^2} = \int_{-\infty}^{\infty} t^2 p_{\mu}(t) dt = \int_0^{24} \frac{t^3}{288} dt = \frac{1}{288} \frac{t^4}{4} \Big|_0^{24} = 288$

$D_{\mu} = E_{\mu^2} - (E_{\mu})^2 = 288 - 256 = 32$